Qualitative:

Quantitive: 1,21

“Before applying a statistical method, we need to understand the nature of the data. And one important way of understanding data is through something called as *“Levels of Measurement”*.

Levels of measurement describe how data values are categorized, ordered, and mathematically compared.

But in simple terms if we want to say

They tell us what we are actually allowed to do with the data

Flow-:

Nominal→ Ordinal→ Interval→ Ratio

1. Nominal Data:

Nominal data consists of categories used only for identification or naming without any inherent order

Nominal basically means names or lables.

Examples:

- Product category
- Payment Method

➔ You can do with nominal data:

- Count
- Percentage (Category contribution)
- Frequency

2. Odrinal Data:

Ordinal data consists of categories that have meaningful order or ranking but the difference between adjacent categories are not necessarily equal or measurable.

“There is an order, but we don’t know whether the gap between the levels is equal.”

Cx Satisfaction

Poor

↓

Average

↓

Good

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↓

Excellent

Rating:

1 Star

2 Stars

3 Stars

4 Stars

5 Stars

Nominal	**Ordinal**
Categories	Categories
No order	Has order
Product type	Satisfaction level
City	Education level
Payment method	Customer rating

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Nominal→ Name

Ordinal → Order

3. Interval Data:

Interval data has ordered values with equal intervals between them, but it does not have a meaningful true zero.

“The difference b/w values is meaningful but zeros does not mean complete absence of the quantity”

Recorded temp of antarctica

4,5,10,0,0,-1,-5

2020, 2nd jan 2020- 9th December 2020 ? year= 0 year

2021,

2022,→ 2021-2022 → 1year

2023

4.Ratio Data

Ratios data has ordered values, equal intervals and a meaningful true zero.

➔ Ratio data has everything interval data has, plus a real zero.

Salary: $0 no salary

Sales: 0 no sales

Weight: 0 no weight

How much distance covered : 0KM

Emp A: 50,000

Emp B: 25,000

➔ I can say Emp A earns twice of Emp B.

Temp A: 20

Temp B: 40

x→ Temp B is twice as hotter as temp A.

Level	Order?	Equal Intervals?	True Zero?
Nominal	❌	❌	❌
Ordinal	✅	❌	❌
Interval	✅	✅	❌
Ratio	✅	✅	✅

Product category→N

Cx satisfaction→ O

Temp in Celcius→ I

Salary → R

Employee ID → We can't do quantitative measure in Emp Id

➢ **The meaning of data matters more than it's appearance.**

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Primary vs Secondary Data

Primary Data:

It is collected first hand by the researcher for a specific research objective

➔ You collect the data yourself for your particular problem.

Secondary Data:

It is data that was previously collected by someone else for another purpose and is reused for analysis.

- Govt Reports
- Research papers
- Company records
- Existing databases
- Industry reports
- Published datasets
- Websites

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Primary	Secondary
Collected first-hand	Already available
Specific to objective	May have been collected for another purpose
More control over collection	Less control
Can be expensive	Usually cheaper to access
Can take more time	Faster to obtain

“ The best source depends on the business objective, quality requirements, cost, time and availability.”

2nd data can be extremely useful if it is :

→Relevant

➔Reliable
➔Recent Enough
➔Collected appropriately

Data Collection Methods:

1. Survey:

A survey collects information from respondents using structured questions.

Usecases:

- CX satisfaction
- Markt research
- Emp Engagement
- Pdct Feedback

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2. Interview:

An interview collects information by asking questions directly to individuals.

10 candidates→ interviewed→ 2 selected

3. Observation:

Observation involves collecting the data by watching and recording behaviour or events.

4. Experiment:

An experiment collects data by deliberately changing a factor under controlled conditions and observing the outcome.

Ecommerce website

V1→ old

V2→ new

Conversion rate→ A/B testing

5. Case Study:

It is an in depth investigation of a specific individual , group, organization, event or situation.

A company studies:

“Why did this startup successfully enter the market?”

6. Existing records:

Data already exists inside the systems

- CRm
- ERP
- Sales Database
- Txn Records
- Employee Records

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Why Do we Sample?

Population → 140cr

States→Capitals→ 1000 people→ collect information

28→ 28000 records→ convenient sampling

“Sampling is the process of selecting a subset of a population to learn about the larger population.”

➔Sampling doesn’t mean randomly taking a few people and hoping for the best.

➔Bad Sample→ Bad conclusion

Common Sampling techniques:

1. Simple Random Sampling:

Every member of the population has an equal chance of being selected.

20,000 std

Randomly select 400

Advantage→ Simple and reduce selection bias when implemented properly.

Limitation→ You need suitable list of the population.

2. Systematic Sampling:

It selects every kth member after choosing a random starting point.

10,000 cx

You need 1,000

I have sampling interval k=10

Cx 7→ 7th, 17th ,27th, 37th,

Advantages-> Easy to implement

Limitations:

If the data contains a repeating pattern related to the interval, the sample can be misleading,

3. Stratified Sampling

It divides the population into meaningful subgroups called strata and select sample from each subgroup

10K std

Divide them department wise

MBA→ 1k

BTECH→5k

MCA→1k

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BBA→ 2k

BA→ 100

PHD→ 500

BCOM→ rest

If we randomly select studs and accidently select very few BA students their perspective may be underrepresented.

Satisfaction helps ensure important groups are presented.

Customers are divided by:

- Region
- Age group
- Customer segment
- Income group

Then samples are taken from each group.

4. Cluster Sampling:

It divides the population into clusters, randomly select clusters, and studies members within the selected clusters**.**

India→ States + UT

Each state and UT represent a cluster

15(UT+State)→ randomly→ study these particular selected states and uts

Stratified→ Divide in groups → take some from the groups

Cluster→ Divide into groups then select some groups

STRATIFIED

Every important group represented

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CLUSTER

Some groups selected

5. Convenience Sampling:

It selects participants because they are easy to access.

Cx feedback→ "Anyone nearby me fill this feedback"

If I select a sample, does that automatically mean my result will be correct?

10M population

Taking 10,000

1. Simple Random Sampling: Selection is random

2. Systematic: Each 3rd person to be selected 3,6,9

3. Stratified: Department:- Company(CS, IT, Mech, Electr)→ Select from each department or group

4. Cluster : Groups already exists: 35 groups → select 100 groups entirely

5. Convenience: Its basically for our convenience

→ Take survey of 10 people→ Take survey from 10 friends.

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Good statistical Analysis = Good data + Good Sample + Good Method

Sampling Bias:

It occurs when the process used to select a sample **systematically** causes the sample to be unrepresentative of the population.

- It means the sample is unfairly selected, so some groups are overrepresented or underrepresented.

1 Lakhs customers (Normal, Average, low value, high value)

Are Cx satisfied with our service?

Analyst surveys only 10k Cx (premium customers)

➔ 90% say they are satisfied.

Can we say? 90% of all are satisfied?

No

Why?

- Company has only included premium
- The company Excluded
  - Regular customers
  - Low value customers
  - New customers
  - Inactive customers.

➔ This is creating a **sampling bias**

If I increase my sample from 10k to 50k premium Cx, have I solved the problem?

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No,

Sample Size is not the same thing as Sample Quality.

Bad Sample and Good Sample

Population 1 Lakh→ Only premium→ 10k Cx (X)

Population 1 Lakh→ premium,low,medium,new→ 10k

Types of Sampling Bias

1. Selection Bias:

Selection bias occurs when some members of the population have different chance of being selected than others because of sampling process.

Example:

A company wants Cx feedback

➔ It surveys the CX who purchased in the last 7 days

➔ The persons who purchased earlier, weekly customers, the customers who stopped purchase will be excluded.

" If we want to understand why customers stopped purchasing, should we only survey active customers?"

➔ No

➔ We need to include the Cx we're are trying to understand.

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2. Non-Response Bias

It occurs when selected individuals do not respond, and the people who respond differ meaningfully from those who do not.

Example:

Company sends a survey to 10,000 cx

Only 2000 respond

Unhappy Cx are less likely to respond

These 2000 responses doesn't mean that the satisfaction is good.

Potential Non- response Bias

3. Coverage Bias:

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It occurs when some members of the target population are not adequately covered or represented by the sampling frame.

Example:

A company wants feedback from **all customers** but conducts survey only with mobile app.

What about the customers who ?

- Does not Use the app
- Does not Own the device
- Uses offline shopping
- Uses website
- Uses older device/ older version of app

They may be underrepresented.

4. Voluntary Response Bias:

It occurs when people choose themselves to participate, and people with stronger opinion are more likely to respond.

Example: A company puts a public survey on it's website

" Tell us how you feel about our service"

Who is more motivated to respond?

- Extremely happy
- Extremely Unhappy

People with neutral opinion may simply ignore it.

Bias	What goes wrong?
Selection Bias	Selection process favors certain groups
Non-response Bias	Selected people don't respond
Coverage Bias	Some population members aren't adequately covered
Voluntary Response Bias	People choose themselves to participate

Suppose—:

- Selected the sample properly

➔ Does that mean my sample result still be different from the whole population?

Yes

Sampling Error:

Sampling error is the difference between a statistic calculated from a sample and the corresponding true population value.

➔ We studied only a sample , so the result from that sample may naturally be different from the true population result.

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Example:

Average customer spending of all customers is $5,000 (50k customers) μ

We select a sample of customers Avg is $ 4,850 $\bar{x}$

Difference is $150

This difference is Sampling error.

2. True Population satisfaction: 60%

Sample result: 57%

Difference is 3 percentage points

The sample result doesn't exactly equal to the population result.

"A properly selected random sample can still produce a statistic that differs from the population value."

Why?

- **A sample contains only a part of the population**

Imagine a population of 1 Lakh, you selected only 1000

Those 1000 won't necessarily have exactly the same characteristics as all 1 lakhs have

Sample Statistic ≠ Population Parameter

➔ A larger, properly selected sample often provides a more stable estimate.

➔ **Increasing sample size can reduce sampling variability, but it does not automatically remove sampling bias.**

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Sampling Bias vs Sampling Error

BIAS: Problem with the sampling process
SAMPLING ERROR: Difference caused by studying a sample

100 average marks =56

40 random students avg = 58

Sampling Bias	Sampling Error
Systematic problem	Natural sample-to-population difference
Related to sample selection/design	Related to using a sample
Can distort results	Causes sample statistics to differ
Larger sample doesn't automatically fix it	Larger proper samples can reduce variability
Example: Premium customers only	Sample average differs from population average

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Q1. Can Sampling Error exists without sampling Bias?

➔ **Yes**
Even a properly selected random sample can differ from the population

Q2. Can bias Exist with a very large sample?

➔ **Yes**
Surveying only premium CX or topper students doesn't make the sample representative of all customers

How to improve the sample quality?

1. **Define the population Clearly**
    a. **Who exactly we are studying**
2. **Use appropriate Sampling Method**
    a. **Simple random**
    b. **Systematic**
    c. **Stratified**
    d. **Cluster**
3. **Ensure that important groups are represented.**
4. **Improve response rate**
    a. **For Surveys**
        i. **Keep questions clear**
        ii. **Keep survey length reasonable**
        iii. **Follow up with non-respondents**
        iv. **Make participation easy**
5. **Check the sampling Frame**
    a. **Ask yourself: ”Does my source actually contain the population I’m trying to study”**

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E commerce customers

1 Million

Management wants to understand:

What is the average monthly spending of our customers?

I. **Population: 1M**
II. **Sample: 10,000**
III. **Sampling Method:** customers are divided by region and the customers are to be selected from each region.--> Stratified Sampling

IV. **Variable:** Monthly Spending

V. **Data type:** Quantitative

VI. **Measurement Level:** Ratio

Why?

Because zero spending has a meaningful interpretation.

VII. **Result:**

a. **Population Average= 4,800**

b. **Sampling Average = 4,730**

c. **Difference= 70**

Potential Sampling error

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Q1. A company has 10 million customers. Why might it use sampling?

Answer: Because studying every customer may be expensive, time-consuming or impractical.

Q2. A company surveys only premium customers to understand all customers. What is the problem?

Answer: Potential selection/sampling bias because the sample may not represent the full customer population.

Q3. If the sample average differs from the population average, does that automatically mean the analyst made a mistake?

Answer: No. A difference can occur naturally because only a sample was studied.

Q4. What is the difference between bias and sampling error?

Answer: Bias is a systematic problem in the sampling process, while sampling error is the difference between a sample statistic and the corresponding population value.

Q5. Can increasing sample size eliminate bias?

Answer: No. A larger sample can still be systematically biased.

Q6. Which sampling method would be useful if you want representation from every region?

Answer: Stratified sampling, assuming region is the relevant grouping variable.

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Q7. What is non-response bias?

Answer: Bias that occurs when selected individuals do not respond and the respondents differ meaningfully from the non-respondents.

Q8. What is the difference between stratified and cluster sampling?

Answer:

Stratified: take samples from each subgroup.

Cluster: select some clusters/groups and study members within those selected clusters.

Q9. What is a parameter?

Answer: A numerical value describing a population.

Q10. What is a statistic?

Answer: A numerical value calculated from a sample.

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